ASSIGNMENT 2

Unit Covered	Unit II — Feature Detection, Segmentation
Suggested Tools	Python (OpenCV, scikit-image), any smartphone camera

1. Photograph (or source) a cluttered tabletop image containing at least three different fruit types with some touching/overlapping, under normal room lighting.
2. Apply edge detection (Canny) and corner detection (Harris or Shi-Tomasi) to the image. Discuss which fruit boundaries are cleanly detected and which are missed, and why (texture, shadow, occlusion).
3. Segment the fruits using at least two different techniques covered in class — choose from watershed, region growing/splitting-merging, or mean-shift — and compare the two segmentation masks side by side.
4. Extract SIFT or HOG descriptors from each segmented fruit region. Pick one apple and one orange, plot their descriptor keypoints, and explain in your own words why these descriptors would (or wouldn't) survive the fruit being rotated 90° on the belt.
5. Apply morphological operations (erosion, dilation, opening, closing) to clean up your segmentation mask where two touching fruits were merged into one blob, and show the before/after.